

Student Registration Form Deployment on AWS EC2 using NGINX, PHP & MariaDB

Project Overview

This project demonstrates deployment of a Student Registration Form web application on an Amazon EC2 instance using Amazon Linux.

The application allows users to submit registration details through a web form, and the entered data is stored successfully in a MariaDB database.

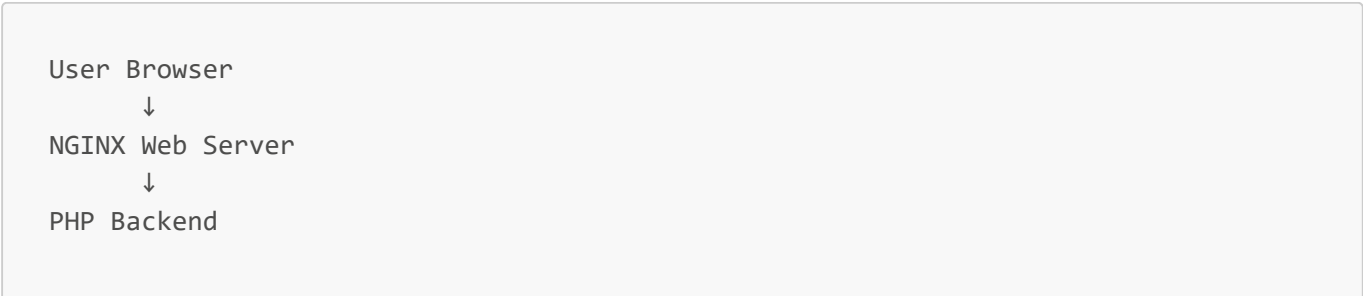
This project was developed to gain practical knowledge in:

- Cloud deployment
- Linux server administration
- Web hosting
- Database integration
- Backend connectivity using PHP

Technologies Used

Technology	Purpose
AWS EC2	Cloud Virtual Server
Amazon Linux	Operating System
NGINX	Web Server
MariaDB 10.5	Database
PHP	Backend
HTML	Frontend

Project Architecture



↓
MariaDB Database

Implementation Steps

Step 1: Create EC2 Instance

Created an Amazon Linux EC2 instance and connected using SSH.

```
ssh -i "first.pem" ec2-user@ec2-13-221-32-203.compute-1.amazonaws.com
```

Step 2: Install Required Packages

Installed:

- NGINX
- MariaDB
- PHP

Command:

```
sudo yum install nginx mariadb105-server php php -y
```

Step 3: Start and Enable Services

Start services:

```
sudo systemctl start nginx mariadb php-fpm
```

Enable services:

```
sudo systemctl enable nginx mariadb php-fpm
```

Check status:

```
sudo systemctl status nginx mariadb php-fpm
```

Step 4: Create Frontend (HTML)

Move to web directory:

```
cd /usr/share/nginx/html/
```

Create signup file:

```
sudo vim signup.html
```

Paste frontend code for Student Signup Form.

Features

- Name
 - Email
 - Website
 - Comment
 - Gender
 - Submit Button
-

Step 5: Configure Database

Login to MariaDB:

```
sudo mysql
```

Set root password:

```
alter user root@localhost identified by 'root';
```

Create database:

```
CREATE DATABASE FCT;
```

Use database:

```
USE FCT;
```

Create table:

```
CREATE TABLE users (  
  id INT PRIMARY KEY AUTO_INCREMENT,  
  name VARCHAR(20),  
  email VARCHAR(100),  
  website VARCHAR(255),  
  gender VARCHAR(6),  
  comment VARCHAR(100)  
);
```

Step 6: Configure Backend (PHP)

Move to web directory:

```
cd /usr/share/nginx/html/
```

Create backend file:

```
sudo vim submit.php
```

Paste PHP backend code to:

- Receive form data
- Connect database
- Insert data into database
- Display successful submission message

Step 7: Install PHP MySQL Connector

Install connector:

```
sudo yum install php8.5-mysqlnd.x86_64 -y
```

Restart services:

```
sudo systemctl restart nginx mariadb php-fpm
```

Step 8: Access Application

Open browser and hit EC2 public IP:

```
http://13.221.32.203/signup.html
```

Student Registration Form opens successfully.

Step 9: Verify Data Entry

Check inserted data:

```
sudo mysql -u root -p
```

Enter password:

```
root
```

Use database:

```
USE FCT;
```

Display records:

```
SELECT * FROM USERS;
```

Data stored successfully in database.

Project Screenshots

Registration Form



Signup Form

Name:

Email:

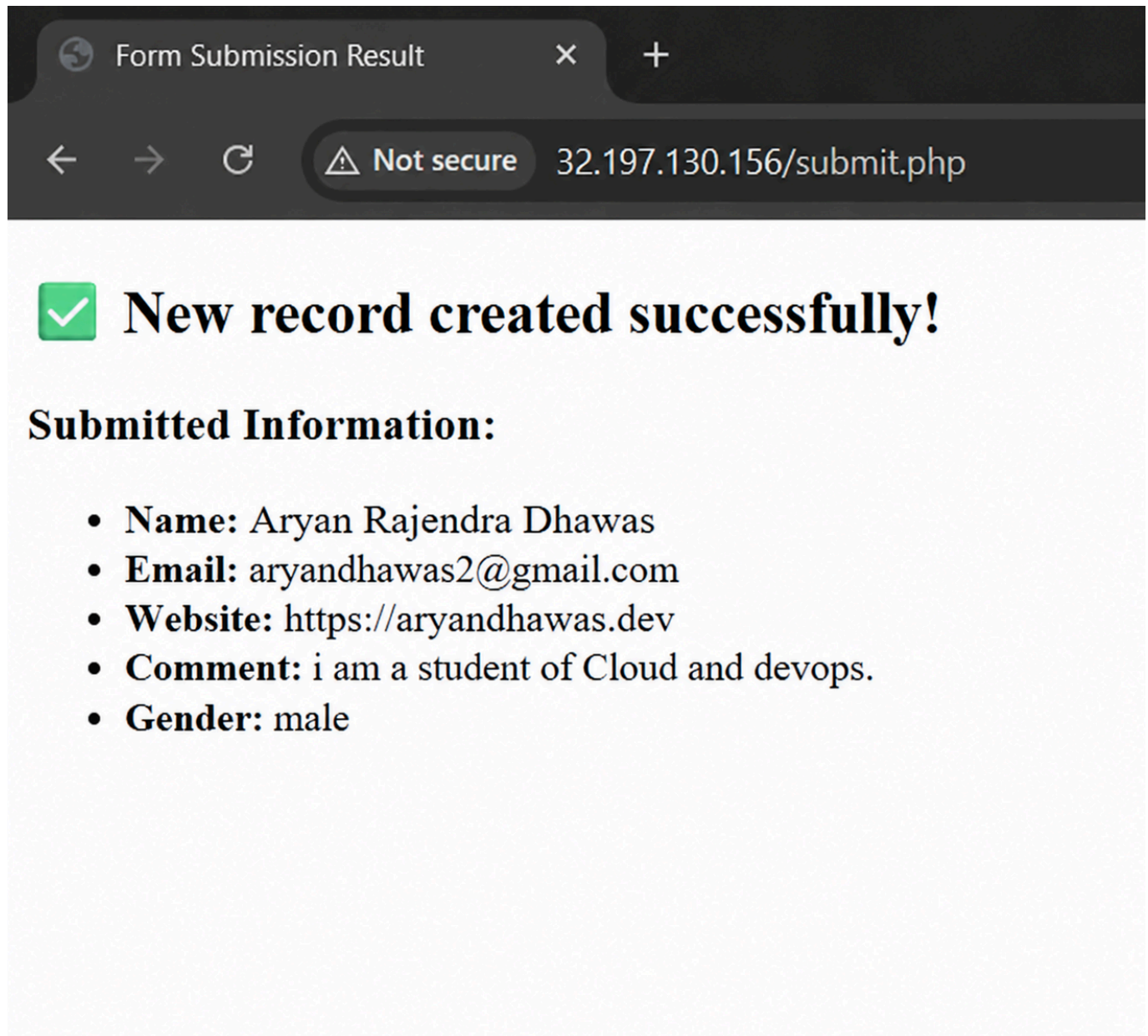
Website:

Comment:

Gender:

- ☐ Female
☐ Male
☐ Other

Form Submission Success



Database Records

```

ERROR 1046 (3D000): No database selected
MariaDB [(none)]> desc databases;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MariaDB server version for
MariaDB [(none)]> show databases;
+-----+
| Database |
+-----+
| FCT      |
| information_schema |
| mysql    |
| performance_schema |
+-----+
4 rows in set (0.002 sec)

MariaDB [(none)]> use FCT;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
MariaDB [FCT]> show tables;
+-----+
| Tables_in_FCT |
+-----+
| users          |
+-----+
1 row in set (0.000 sec)

MariaDB [FCT]> desc users;
+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+
| id    | int(11) | NO | PRI | NULL | auto_increment |
| name  | varchar(20) | YES | | NULL | |
| email | varchar(100) | YES | | NULL | |
| website | varchar(255) | YES | | NULL | |
| gender | varchar(6) | YES | | NULL | |
| comment | varchar(100) | YES | | NULL | |
+-----+
6 rows in set (0.001 sec)

MariaDB [FCT]> select * from users;
+-----+
| id | name | email | website | gender | comment |
+-----+
| 1 | aryan dhawas | aryandhawas2@gmail.com | http://54.80.101.161/signup.html | male | bye! |
| 2 | sujit more | sujitmored23@gmail.com | http://32.197.130.156/signup.html | male | i am a student of Cloud and devops. |
+-----+
2 rows in set (0.000 sec)
MariaDB [FCT]>

```

Database Records

AWS EC2 Dashboard

The screenshot shows the AWS Global View of the EC2 console. On the left, the 'Instances' section is expanded. The main area displays a table of instances:

Name	Instance ID	Instance state	Instance type	Status check	Alarm status
amazon linux db	i-0dc9a0aeb1cbdc3cc	Stopping	t3.micro	...	View alarms
my_portfolio	i-070318c4af7b427ab	Stopped	t3.micro	...	View alarms
Webserver	i-0742fd344d9b0ea9f	Stopped	t3.micro	...	View alarms
ubuntu	i-09418db290c1e2c68	Stopped	t3.micro	...	View alarms

Below the table, the details for the selected instance 'i-070318c4af7b427ab (my_portfolio)' are shown. The 'Details' tab is active, displaying various configuration options like Status and alarms, Monitoring, Security, Networking, Storage, and Tags.

Project Output

- Student registration form deployed successfully
- Form accessible using EC2 public IP
- User data stored in MariaDB database
- Full cloud deployment completed successfully

Learning Outcomes

Through this project I learned:

- AWS EC2 deployment
 - Linux server management
 - NGINX web server configuration
 - MariaDB database setup
 - PHP backend integration
 - Full-stack cloud deployment
 - Troubleshooting services
-

Problems Faced During Deployment

During deployment, some common issues were faced:

- Database connection errors
- PHP connector installation problems
- Service restart issues
- Linux command mistakes
- MariaDB syntax errors

These issues were solved using troubleshooting commands and service status checking.

Useful Commands

Show Databases

```
SHOW DATABASES;
```

Use Database

```
USE FCT;
```

Show Tables

```
SHOW TABLES;
```

Describe Table

```
DESC users;
```

Display Records

```
SELECT * FROM users;
```

Future Improvements

Future improvements planned for this project:

- Add CSS styling
- Add Bootstrap UI
- Add Login Authentication
- Add Password Encryption
- Add HTTPS SSL Certificate
- Deploy using Load Balancer
- Docker Containerization
- CI/CD Pipeline

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Summary

This project is a simple cloud-based Student Registration Form application developed using AWS EC2, NGINX, PHP, and MariaDB.

The project provided practical experience in cloud hosting, Linux server management, backend integration, and database handling in a real-time environment.